[EA&B] CDXR: A Contract-Grounded Evaluation Architecture for Tabular Leakage Repair

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Abstract

Leakage repair is usually evaluated as a feature-detection problem, even though four different questions determine whether an intervention is defensible: Is a field invalid at prediction time? Can a blind statistic locate it? Does the learner exploit it? Does removing it improve a boundary-valid reference beyond what equal-cost deletion would achieve? We introduce *CDXR*, a contract-grounded evaluation architecture that makes construction validity (C), detectability (D), exploitability (X), and repair response (R) separate, auditable objects. CDXR enforces semantic precedence, oracle-isolated policy selection, matched intervention cost, paired downstream evaluation, and task-level uncertainty. We instantiate the architecture in LeakBench-Tab, a frozen registry with 27,500 measurement cells spanning 20 designed tasks, 11 leakage mechanisms, five strengths, five injection seeds, and five learners. A 709,500-row repair ledger then compares training-side mutual-information removal with 20 matched random-removal realizations per key. Under the primary 20% encoded-column contract, the LR repair advantage is +0.043 (95% task-cluster interval [+0.004, +0.078]). A corroborative post-LR extension, whose learner pair, budget, seeds, and estimand were frozen before its outcomes, gives +0.055 for RF ([+0.025, +0.082]) and +0.056 for LightGBM ([+0.028, +0.082]) under disclosed 100-estimator baseline refits. The architecture exposes why the aggregate is conditional: low-opportunity strata are negative, M09 is a positive structured counterexample, semantic-group cost moves the recomposed overall interval across zero, and five fixed natural cases provide mixed evidence. CDXR therefore converts leakage repair from an uncalibrated deletion heuristic into a bounded, falsifiable evaluation claim.

Keywords

data leakage, repair evaluation, information contracts, matched-cost governance, tabular machine learning

1 Introduction

Tabular leakage is not one statistical defect. A field can violate the prediction-time contract while showing weak marginal association; a valid field can be strongly predictive; and a learner can exploit an invalid interaction that a univariate detector barely ranks. Nevertheless, leakage audits often compress semantic invalidity, statistical suspicion, and observed performance inflation into one label [8, 9]. That compression becomes consequential when an analyst moves from diagnosis to deletion.

Deletion is an intervention on a learned representation. Removing a suspicious field can eliminate invalid information, discard legitimate signal, or merely regularize the learner. A comparison against the unmodified table cannot separate these effects. A comparison against an oracle mask instead measures perfection, not whether a blind ranking adds value. The missing control is an equal-cost intervention that removes the same amount of representation without using the ranking. Only then does the contrast answer a governance question: did the policy improve the boundary-valid reference beyond deletion alone?

Existing work supplies important pieces of this problem. Semantic leakage frameworks emphasize the prediction boundary and scientific validity [4, 16]; static and empirical audits search for contamination at code or data level [11, 19]; and feature-selection methods rank predictive variables. None of these pieces alone specifies an evaluation contract for a budget-limited repair policy. The contract must declare which fields are invalid before scoring, hide that declaration from non-oracle selectors, hold intervention cost fixed, and preserve the task hierarchy during inference.

We introduce **CDXR**, a contract-grounded architecture for this missing evaluation layer. CDXR separates four objects: construction validity (C), blind detectability (D), learner-conditional exploitability (X), and matched-cost repair response (R). The architecture does not propose mutual information as a new detector. Instead, it specifies the interfaces and controls required to turn any candidate selector into a falsifiable repair claim. LeakBench-Tab is the controlled instantiation: eleven audited mechanisms perturb a fixed prediction boundary, and a representative MI policy is evaluated against matched random deletion.

The architecture yields four research questions. **RQ1** asks whether D and X coincide across mechanisms. **RQ2** asks whether a blind ranking adds repair value beyond equal-cost deletion across learners and budgets. **RQ3** asks how repair opportunity and task archetype condition that response. **RQ4** asks whether the conclusion survives changes in the cost unit and application context. Each question targets one CDXR interface rather than treating all successful cells as interchangeable evidence.

Contributions. This paper makes three contributions.

- We formalize CDXR and its four protocol invariants: semantic precedence, oracle isolation, matched intervention cost, and paired reference evaluation. The repair layer returns downstream distance, contamination recall, and legitimate retention rather than a detector score alone.
- We instantiate CDXR in a frozen factorial registry with 27,500 measurement cells and a 709,500-row multi-seed repair ledger. The design distinguishes learner, mechanism, opportunity, archetype, and cost sensitivity while keeping the controlled task as the independent unit.

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- We establish a bounded empirical result: MI-guided removal has a positive primary encoded-cost repair advantage for LR, corroborated by separately frozen RF and LightGBM extensions. Low-opportunity strata, the sparse archetype, semantic-group recomposition, and NYC311 expose where that statement narrows or fails.

2 The CDXR Architecture

2.1 Contract Schema and Role-Separated Interfaces

A CDXR evaluation begins with an explicit contract, not a feature score. For task q, the minimum contract is

$$\mathcal{K}_q = (p_q, h_q, \mathcal{G}_q, \rho_q, a_q, \mathcal{M}_q^{\text{pol}}, \kappa_q, \Pi_q, \ell, \mu). \quad (1)$$

Here p_q and h_q declare the prediction event and horizon; $\mathcal{G}_q$ is the set of semantic fields; ρ_q maps each semantic field to its encoded columns; and a_q records availability and lineage at p_q. The remaining entries declare policy-visible metadata $\mathcal{M}_q^{\text{pol}}$, intervention cost κ_q, split and independent-unit plan Π_q, learner ℓ, and metric μ. The hidden oracle mask $\mathcal{L}_q$ is induced from (p_q, h_q, ρ_q, a_q) before any diagnostic or learner outcome is inspected.

CDXR separates three roles. A selector receives training data, its declared metadata, and a budget, but not the oracle mask or held-out outcomes:

$$\phi_r : (X_q^{\text{tr}}, y_q^{\text{tr}}, \mathcal{M}_q^{\text{pol}}, k) \longmapsto \mathcal{R}_{q,r,k}. \quad (2)$$

An evaluator fits the named learner on strict, full, and governed views under Π_q and returns their held-out scores. Only after $\mathcal{R}_{q,r,k}$ is frozen may a verifier read $\mathcal{L}_q$ and emit the claim certificate

$$\Gamma = (S, u, \widehat{\Delta}_R, \mathrm{CI}, \mathrm{Rec}, \mathrm{Ret}, P, z), \quad (3)$$

where S is the claim scope, u the cost unit, P the provenance binding, and z the claim status. The verifier returns INVALID if contract completeness, role access, cost matching, pairing, or provenance fails. Given a valid protocol, a prespecified positive claim is SUPPORTED only when its task-level interval excludes zero in the declared direction; otherwise it is UNSUPPORTED. A claim is NARROWED when a declared mechanism, cost, or context sensitivity changes its support boundary, while discordant fixed cases without a population estimand are MIXED. These gates map directly to the artifact's machine-built claim state. A lineage-aware, time-aware, or group-aware policy plugs into the same selector by changing $\mathcal{M}_q^{\text{pol}}$ and κ_q; it may not change which role can read $\mathcal{L}_q$.

2.2 Construction, Detectability, and Exploitability

Let $q = (t, m, a, s)$ index a controlled cell by task t, contamination mechanism m, strength a, and injection seed s. Its encoded feature universe is $\mathcal{F}_q$, and the construction contract identifies the contaminated subset $\mathcal{L}_q \subseteq \mathcal{F}_q$. The strict and permissive views are therefore

$$\mathcal{F}_q^{\text{strict}} = \mathcal{F}_q \setminus \mathcal{L}_q, \qquad \mathcal{F}_q^{\text{full}} = \mathcal{F}_q. \quad (4)$$

Each task declares an information boundary at prediction time t_p. For encoded field j, construction validity is

$$C_{qj} = \mathbf{1}\big[j \in \mathcal{L}_q\big] = \mathbf{1}\big[j \text{ is unavailable or target-derived at } t_p\big]. \quad (5)$$

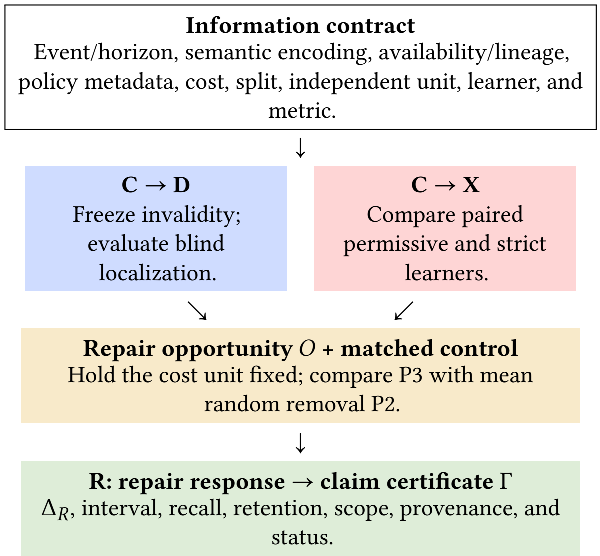

Figure 1: The CDXR architecture. The information contract defines C before any score or learner is inspected. D and X are measured independently. The initial strict–full distance defines repair opportunity; R is evaluated against an equal-cost negative control. The diagram shows interfaces rather than a production deployment architecture.

C_{qj} is fixed by task semantics and mechanism construction. It is not inferred from association, attribution, or downstream performance.

An oracle-blind diagnostic assigns score s_{qj} without reading C_{qj} or the test rows. We measure localization with average precision and normalize it by the contaminated field prevalence π:

$$\pi_q = \frac{|\mathcal{L}_q|}{|\mathcal{F}_q|}, \qquad D_q = \frac{\mathrm{AP}_q - \pi_q}{1 - \pi_q}. \quad (6)$$

The primary diagnostic is mutual information computed on training rows with a fixed random state. D is conditional on the diagnostic and representation.

For learner ℓ, exploitability is the paired AUROC difference

$$X_{q\ell} = A_{q\ell}^{\text{full}} - A_{q\ell}^{\text{strict}}, \quad (7)$$

where ℓ indexes the downstream learner and A denotes test AUROC. The strict view removes construction-labeled contaminated fields; the permissive view retains them. Positive X measures performance inflation under this protocol. An interval containing zero is reported as an imprecise or unsupported directional effect, not as evidence of equivalence.

2.3 Repair Opportunity and Response

For a policy r and removal budget k, let $\mathcal{R}_{q,r,k} \subseteq \mathcal{F}_q$ be its selected removal set. The governed feature view is

$$\mathcal{F}_{q,r,k}^{\text{gov}} = \mathcal{F}_q \setminus \mathcal{R}_{q,r,k}. \quad (8)$$

Let A_s, A_f, and A_g denote strict, full, and governed test AUROC for a fixed q and learner. Their initial distance defines the available repair opportunity

$$O_{q\ell} = |A_{q\ell}^{\text{full}} - A_{q\ell}^{\text{strict}}| = |X_{q\ell}|. \quad (9)$$

Opportunity is necessary for a large score repair but is not sufficient for localization: a policy must still identify a useful intervention. We measure strict distance reduction (SDR) as

$$\mathrm{SDR}_{q\ell rk} = O_{q\ell} - |A^{\mathrm{gov}}_{q\ell rk} - A^{\mathrm{strict}}_{q\ell}|. \tag{10}$$

Positive SDR means governance moves the evaluated learner closer to the strict reference. This symmetric distance avoids treating negative residual harm as automatic success. We also report contaminated field recall and legitimate retention:

$$\mathrm{Rec}_{qrk} = \frac{|\mathcal{R}_{q,r,k} \cap \mathcal{L}_q|}{|\mathcal{L}_q|}, \tag{11}$$

$$\mathrm{Ret}_{qrk} = 1 - \frac{|\mathcal{R}_{q,r,k} \setminus \mathcal{L}_q|}{|\mathcal{F}_q \setminus \mathcal{L}_q|}. \tag{12}$$

The R layer is therefore the vector

$$R_{q\ell rk} = \left(\mathrm{SDR}_{q\ell rk}, \mathrm{Rec}_{qrk}, \mathrm{Ret}_{qrk}\right). \tag{13}$$

For candidate P3 and random control P2 with governance realization ω, the primary scalar contrast is the repair advantage

$$\Delta_{R,q\ell k} = \mathrm{SDR}_{q\ell,\mathrm{P3},k} - \mathbb{E}_{\omega}\left[\mathrm{SDR}^{(\omega)}_{q\ell,\mathrm{P2},k}\right]. \tag{14}$$

These R outcomes may read C only after a policy has selected fields; a non-oracle selector cannot.

2.4 Architectural Invariants

Four invariants distinguish CDXR from ordinary feature selection. *Semantic precedence* requires $\mathcal{L}_q$ to be determined by the prediction-time contract rather than by a score threshold. A feature can be invalid with low marginal association, or valid with high association; neither case is contradictory. *Oracle isolation* requires the selection map $\mathcal{R}_{q,r,k} = \phi_r(X^{\mathrm{train}}_q, y^{\mathrm{train}}_q, k)$ to exclude $\mathcal{L}_q$ for non-oracle policies. The mask enters only after $\mathcal{R}_{q,r,k}$ is frozen, when recall and retention are evaluated.

Cost matching declares a cost map $\kappa^{(u)}_q : 2^{\mathcal{F}_q} \to \mathbb{N}$ for representation unit u and requires

$$\kappa^{(u)}_q(\mathcal{R}_{q,\mathrm{P2},k}) = \kappa^{(u)}_q(\mathcal{R}_{q,\mathrm{P3},k}) = k. \tag{15}$$

The primary unit is an encoded column. Removing all eight M09 source indicators therefore costs eight units, while removing one numeric target proxy costs one. A frozen sensitivity replaces u with a semantic feature group. CDXR treats this unit as part of the intervention contract rather than a formatting detail.

Paired reference evaluation keeps the task, split, mechanism, strength, injection seed, and learner seed fixed when strict, full, and governed AUROC are compared. SDR is bounded by $[-1, 1]$ because it is a difference of two AUROC distances. Positive SDR indicates movement toward the strict score, but not recovery of the strict feature set: a policy may reach a similar AUROC after removing the wrong fields. Recall and retention are reported alongside SDR precisely to reveal that ambiguity. Likewise, negative SDR identifies movement away from the strict reference but does not by itself distinguish residual contamination from over-removal of legitimate signal.

These invariants define the scope of a repair claim. A policy is useful in CDXR only when it improves the paired distance under equal removal cost and its localization trade-off remains visible. The formulation does not assert that the strict model is optimal for a business objective, only that it is the evaluation reference consistent with the declared information boundary.

3 Controlled Instantiation and Design

3.1 Controlled Registry

The controlled registry contains 20 deterministic panel tasks spanning five data-generating archetypes. Each task has a chronological 60/20/20 split and separates generator seeds from injection seeds. Eleven mechanisms cover direct target copies (M01), noisy proxies (M02), nonlinear transforms (M03), post-outcome aggregates (M04), temporal look-ahead (M05), redundant clusters (M06), sparse subgroup leakage (M07), entity leakage (M08), source leakage (M09), mixed legitimate and contaminated components (M10), and graph-mediated leakage (M11). Categories are assigned by construction: M01, M02, M06, and M10 are simple; M03, M07, and M11 are boundary mechanisms; M04, M05, M08, and M09 are structured. Categories are not revised after observing results.

The five base archetypes are linear, interaction, nonlinear, sparse, and drifting binary prediction tasks. Each contains entity and time structure and between 12 and 24 original numeric fields. Clean signal combines row-level covariates with entity effects and a periodic time component; the archetype then changes the functional form rather than the evaluation contract. Four tasks instantiate each archetype. Task generator seeds are deterministic functions of the panel identifier, so changing an injection or learner seed cannot regenerate the clean table. All tasks are ordered by timestamp before the 60/20/20 split, and injection is verified not to alter the frozen row indices.

This panel is designed for controlled variation rather than realism by volume. It supplies repeated task structure in which semantic contracts and injected features are exactly auditable. The five natural cases introduced below serve a different purpose: they test whether boundary-specific contamination remains visible in selected real tables, but they do not replace the controlled panel as the inferential unit.

The core matrix crosses 20 tasks, 11 mechanisms, five strengths, five injection seeds, and five learners, yielding 27,500 paired cells. The learners are logistic regression, random forest [1], CatBoost [15], LightGBM [10], and TabM [6]. Preprocessing and diagnostic fitting use training data only. The design evaluates conventional linear, bagged-tree, boosted-tree, and neural tabular families without treating model family differences as causal capacity effects.

3.2 Mechanism Families and Strength Semantics

The registry varies how invalid information becomes available to a learner. The *simple* family contains fields whose encoded columns directly expose or redundantly repeat target information. M01 copies the current label with a strength-dependent flip rate; M02 adds noise to a centered target proxy; M06 creates a correlated cluster whose size increases across strength levels; and M10 pairs one legitimate copy of an original field with one contaminated target proxy. M10 is deliberately mixed: the strict view removes only its contaminated component and retains the legitimate component, preventing the oracle reference from benefiting merely by deleting an entire injected group.

The *structured* family makes validity depend on relations not represented by a single contemporaneous field. M04 aggregates labels from a strictly future window; M05 introduces a configurable look-ahead offset; and M08 computes a shrunken, strictly future label rate within entity. All three exclude the current row's own label. M09 assigns one of eight source categories with outcome-dependent source probabilities and represents the source as a complete one-hot block. It does not use target-rate encoding, but the source identity is deployment-invalid under the declared construction. Consequently, its detectability and removal cost are representation-conditional.

The *boundary* family stresses cases between those poles. M03 multiplies a target-derived signal by the side of a threshold on a legitimate field, hiding marginal association behind an interaction. M07 exposes a target proxy only within a sampled subgroup, with subgroup prevalence varying by strength. M11 distributes target-derived signal across a connected component whose size also varies with strength. These mechanisms are not claimed to enumerate all forms of leakage. They are controlled probes for directness, sparsity, interaction, redundancy, time, entity, source, and group structure.

Strength labels S1–S5 correspond to values 0.2, 0.4, 0.6, 0.8, and 1.0, but the operational parameter is mechanism-specific. Strength scales signal for direct proxies, changes subgroup coverage for M07, and changes group size for M06 and M11. It therefore defines an ordered intervention within a mechanism, not a common physical scale across mechanisms. Cross-mechanism comparisons average the frozen five-level profile; they do not interpret S3 in one mechanism as quantitatively equivalent to S3 in another.

3.3 Diagnostics and Learner Protocol

The primary diagnostic assigns each encoded field a univariate MI score using training rows and a fixed random state. Localization is evaluated after scoring with normalized average precision, which adjusts for the mechanism-specific prevalence of contaminated encoded columns. Average precision is appropriate for this imbalanced ranking problem because most fields are legitimate and the relevant operating region is concentrated near the top of the ranking [17]. We also record top-five contaminated-field recall as an operational summary, but the normalized AP metric supplies D in the C/D/X analysis.

A frozen sensitivity suite evaluates three alternatives: absolute Pearson correlation on training rows, absolute coefficients from standardized logistic regression, and validation-set permutation importance from a training-fitted random forest. None receives the contamination mask before producing scores. The four-method suite contains 22,000 successful diagnostic cells. Because the protocol did not freeze simultaneous paired method-comparison intervals, these results diagnose method dependence but do not support an inferential ranking of detectors. In particular, a retrospectively selected best method is an optimistic labeled-benchmark summary, not a deployable selector.

The downstream panel fixes five named implementations. LR uses standardized features and logistic regression; RF uses 250 trees; CatBoost and LightGBM use validation-only early stopping; and TabM uses the official pinned package with 32 ensemble members, training-fitted standardization, deterministic execution, and validation-loss checkpointing. Test labels are used only to compute final AUROC. Missing packages or accelerators fail a cell rather than triggering a fallback under another model identity. The completed canonical matrix contains all 27,500 planned model cells, so no failed-cell exclusion enters the reported means.

3.4 Repair Policies and Negative Control

The R experiment uses the same 5,500 combinations of task, mechanism, strength, and injection seed. LR is evaluated at all four budgets; RF and LightGBM test the primary 20% budget. LR is the primary repair panel. The post-LR B2 extension uses separately frozen repair factories: a 100-tree RF and a 100-estimator CPU LightGBM, both with the recorded training seed and without the measurement panel's RF-250 or validation-early-stopping configuration. They are therefore corroborative learner interventions rather than bitwise continuations of the five-learner measurement matrix. P0 retains all fields. P1 removes every construction-labeled field and defines the strict oracle reference. P2 removes k fields uniformly at random under 20 frozen governance seeds. P3 removes the k fields with the largest training-side MI scores. If $\omega \in \Omega$ denotes one of $|\Omega| = 20$ frozen P2 realizations and I_{qj}^{train} denotes the training-only score, then

$$\mathcal{R}^{(\omega)}_{q,\text{P2},k} \sim \text{Unif}\{R \subseteq \mathcal{F}_q \,:\, |R| = k\}, \tag{16}$$

$$\mathcal{R}_{q,\text{P3},k} = \text{TopK}_{j\in\mathcal{F}_q}\left(I_{qj}^{\text{train}}; k\right). \tag{17}$$

For a nominal budget p of 1%, 5%, 10%, or 20% and $d_q = |\mathcal{F}_q|$ encoded fields, both policies use

$$k_q(p) = \max\Big(1, \text{round}(pd_q)\Big). \tag{18}$$

Thus the comparison satisfies the cellwise cost identity

$$|\mathcal{R}^{(\omega)}_{q,\text{P2},k}| = |\mathcal{R}_{q,\text{P3},k}| = k_q(p). \tag{19}$$

P2 and P3 therefore remove exactly the same number of fields for each key and budget. The implementation estimates Eq. 14 by averaging P2 within key before task-cluster resampling. Nominal fractions that collapse to the same integer k reuse the same fitted result and retain both labels in the ledger.

The clean runner loads frozen bundles, verifies bundle and split hashes, and does not inject, split, or mutate data. Non-oracle selectors never receive the contamination mask. Strict and full baselines are shared only within an identical task, mechanism, strength, and seed key. The final revision contains 709,500 successful policy rows across 5,500 keys per learner, with complete selection hashes and no duplicate run identifiers.

3.5 Natural-Case Audit

Five selected public or locally mirrored tables instantiate concrete prediction boundaries: call completion for Bank Marketing, loan origination for Lending Club, scheduled departure for U.S. flights, inspection completion for Chicago food inspections, and complaint creation for NYC 311. For each task, the strict view removes fields declared unavailable at that boundary and the permissive view retains them. The harm audit crosses four CPU learner families with three seeds, producing 60 paired cells. A separate LR governance sensitivity uses the 20% retained-feature budget, three training seeds, and 20 P2 governance seeds, producing 315 policy cells. The five cases are selected rather than sampled, so both analyses remain descriptive fixed-case evidence.

Categorical preprocessing is fitted on training rows. Categories absent from training map to a reserved unknown value, missing categories receive a separate value, and date strings that had previously been globally encoded are dropped. This policy prevents validation or test vocabularies from defining the training representation. The case selection is intentionally heterogeneous in table width and contamination count, but it is not random and does not define a population of application datasets. One Lending Club source remains a local mirror whose redistribution license is unresolved; the paper reports its boundary-specific result but excludes the raw file from the public artifact.

3.6 Freeze and Amendment Discipline

Confirmatory execution is separated from protocol development. The canonical manifest binds task bundles, split hashes, configuration, runner code, model identity, and source result tables. Pilot namespaces use disjoint task seeds. Once a confirmatory protocol is frozen, a change requires an explicit amendment and replacement of the affected cells rather than silent in-place repair.

The canonical table incorporates four documented corrections. M04, M05, and M08 use strictly future definitions that satisfy their construction contracts. M10's strict mask retains its legitimate injected field. MI uses a fixed random state on training data for the affected diagnostic cells, and natural-case preprocessing is training-fitted. Superseded rows are retained for provenance but excluded from canonical claim generation. These amendments narrow the evidence to the final protocol and prevent earlier, semantically different outputs from being averaged into the paper results. The B2 chronology is part of the evidence tier. The LR-only limitation was visible first. The dated revision protocol then froze RF and LightGBM, the 20% budget, 20 governance seeds, model factories, and the within-key P3-minus-mean-P2 estimand before either cross-learner outcome was observed. After the RF outcome, LightGBM moved from a failed GPU attempt to CPU before any completed LightGBM result; its model class, estimator count, seeds, budget, and estimand were unchanged. B2 is therefore a prespecified post-LR extension, not part of the original measurement protocol.

A later audit found that the B2 runner re-fitted strict and full baselines with its 100-estimator repair factories instead of numerically reusing the earlier canonical RF-250 and early-stopped LightGBM fits. This post-run deviation is bound in the revision manifest and claim state. It preserves within-row pairing and the frozen P3–P2 estimand, but it restricts RF and LightGBM to corroborative within-run contrasts rather than baseline-continuity evidence.

3.7 Statistical Analysis and Claim Scope

The controlled task is the independent unit. Core measurement intervals use a 20,000-draw hierarchical bootstrap that samples tasks and then injection seeds. Mechanism tests use task-level sign flips with Holm correction over the eleven mechanisms. The three category contrasts form a separate Holm family. The prespecified simple-minus-structured contrast receives the core binary support decision; individual mechanism profiles remain descriptive.

For clarity, let $\mathscr{V}$ denote the learner set, and let $\mathscr{M}_{\mathrm{si}}$ and $\mathscr{M}_{\mathrm{st}}$ denote the simple and structured mechanism families. The category-level harm for task t and seed s is

$$\bar{X}_{t,s,c} = \frac{1}{|\mathscr{V}||\mathscr{M}_c||\mathscr{A}|} \sum_{\ell\in\mathscr{V}} \sum_{m\in\mathscr{M}_c} \sum_{a\in\mathscr{A}} X_{(t,m,a,s),\ell}, \tag{20}$$

so the prespecified contrast is

$$\hat{\Gamma}_{\mathrm{si-st}} = \frac{1}{|\mathscr{T}||\mathscr{S}|} \sum_{t\in\mathscr{T}} \sum_{s\in\mathscr{S}} \left(\bar{X}_{t,s,\mathrm{si}} - \bar{X}_{t,s,\mathrm{st}}\right). \tag{21}$$

Writing the seed-averaged paired contrast as e_t, its exact two-sided task-level sign-flip value is

$$p_{\mathrm{flip}} = 2^{-|\mathscr{T}|} \sum_{\epsilon\in\{-1,+1\}^{|\mathscr{T}|}} \mathbf{1}\left[\left|\frac{1}{|\mathscr{T}|}\sum_t \epsilon_t e_t\right| \geq \left|\frac{1}{|\mathscr{T}|}\sum_t e_t\right|\right]. \tag{22}$$

For core confidence intervals, each bootstrap draw samples 20 tasks with replacement and then samples five injection seeds with replacement inside each selected task. All strengths, mechanisms, and learners associated with a drawn task–seed pair remain together. This preserves the pairing that defines $X_{q\ell}$ and prevents the 27,500 model cells from acting as independent replicates. The simple-minus-structured interval is the 2.5th and 97.5th percentiles of 20,000 hierarchical draws. Its exact sign-flip test first averages seeds within task, enumerates all 2^{20} task-sign assignments, and uses the two-sided absolute mean. Holm adjustment is applied jointly to the three declared family contrasts. The eleven mechanism tests form a different Holm family; their profiles remain descriptive regardless of adjusted value.

Repair differences are first averaged within each of the 20 controlled tasks. Let $\overline{\mathrm{SDR}}^{\Omega}_{q\ell,\mathrm{P2},k}$ denote the within-key mean over the 20 P2 realizations. With $\mathscr{M}$, $\mathscr{A}$, and $\mathscr{S}$ denoting the mechanism, strength, and injection-seed sets, the learner-specific task effect is

$$d_{t\ell p} = \frac{1}{|\mathscr{M}||\mathscr{A}||\mathscr{S}|} \sum_{m\in\mathscr{M}} \sum_{a\in\mathscr{A}} \sum_{s\in\mathscr{S}} \Big(\mathrm{SDR}_{q\ell,\mathrm{P3},k_q(p)} - \overline{\mathrm{SDR}}^{\Omega}_{q\ell,\mathrm{P2},k_q(p)}\Big), \tag{23}$$

and the registry-level estimand is the equal-task average

$$\widehat{\Delta}_{R,\ell p} = \frac{1}{|\mathscr{T}|} \sum_{t\in\mathscr{T}} d_{t\ell p}. \tag{24}$$

For a mechanism family c with member set $\mathscr{M}_c$, replacing $\mathscr{M}$ by $\mathscr{M}_c$ in Eq. 23 gives

$$\widehat{\Delta}_{R,\ell pc} = \frac{1}{|\mathscr{T}|} \sum_{t\in\mathscr{T}} d_{t\ell pc}. \tag{25}$$

We assess these estimands with a paired 10,000-draw cluster bootstrap over tasks. For draw b, tasks are sampled with replacement and the complete task effect is carried with each sampled index:

$$\widehat{\Delta}^{*(b)}_{R,\ell p} = \frac{1}{|\mathscr{T}|} \sum_{i=1}^{|\mathscr{T}|} d_{T_i^{*(b)},\ell p}, \quad T_i^{*(b)} \overset{\mathrm{iid}}{\sim} \mathrm{Unif}(\mathscr{T}). \tag{26}$$

We report the observed mean task difference, percentile interval, and the fraction of bootstrap draws above zero:

$$\begin{aligned} \mathrm{CI}_{0.95} &= \left[Q_{0.025}(\widehat{\Delta}^{*}_{R,\ell p}), Q_{0.975}(\widehat{\Delta}^{*}_{R,\ell p})\right], \\ \widehat{P}_{\mathrm{P3>P2}} &= \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}[\widehat{\Delta}^{*(b)}_{R,\ell p} > 0]. \end{aligned} \tag{27}$$

Learner interactions are direct paired contrasts of task effects, not comparisons of whether two separate intervals contain zero. Gap quartiles and archetype rows reuse the same LR task-cluster procedure as declared sensitivities. Semantic-cost and fixed-natural-case summaries use 5,000 draws; the natural analysis also reports the exact two-sided sign-flip value over its five fixed case effects. The bootstrap fraction is not a p-value. Because tasks form a designed registry rather than a sample from a defined task population, all intervals quantify stability to registry reweighting. They do not support population inference to arbitrary tabular datasets.

4 CDX Measurement Results

4.1 X: Construction Family Changes Exploitability

The prespecified simple-minus-structured contrast in paired AUROC harm was 0.164 (95% hierarchical bootstrap CI [0.150, 0.178]; Holm-adjusted exact task-level sign-flip $p = 5.72e - 06$). The direction was positive across all five core learner families. This category average does not imply that every structured mechanism is harmless: M09 is a structured counterexample with mean harm 0.235.

The learner-family consistency is descriptive rather than a second support decision. LR, bagged trees, boosted trees, and TabM differ in their ability to represent interactions and partitions, but each yielded a positive simple-minus-structured average. This observation reduces the likelihood that the pooled direction is produced by one learner family alone. It does not show that learner capacity causes the category contrast, because hyperparameters, regularization, and representation are fixed components of the named model implementations rather than randomized treatments.

The mechanism rows further qualify the category summary. M01 and M02 show the expected vulnerability to explicit target proxies, while redundant M06 and mixed M10 yield still larger mean harm. M04 and M05 are construction-invalid despite estimates close to zero: their future-window values can be weakly useful in this particular generator and representation. M09 demonstrates the opposite possibility. Its source categories are structurally defined, yet the complete one-hot representation makes the source shift exploitable. Therefore "structured" describes how invalid information is organized, not a guarantee of low statistical association or low downstream harm.

4.2 D Does Not Determine X

Mechanism profiles show why a single leakage severity score is insufficient. M03 combines low primary-MI detectability (D=0.079) with substantial harm (X=0.217). M08 has intermediate detectability (D=0.356) and an imprecise small mean harm (X=0.003, CI spanning zero). M09 is both more detectable and strongly exploitable in the fixed one-hot source representation. These rows are descriptive profiles, not post hoc thresholded claims.

The diagnostic sensitivity suite makes the conditioning on D concrete. Across all mechanisms, the aggregate ordering of the four diagnostics differs from their ordering on individual mechanisms. M03 is the clearest example: its normalized AP ranges from 0.035 under absolute correlation to 0.602 under RF permutation importance, whereas primary MI yields 0.079. M04 and M05 remain weakly localized under all four evaluated diagnostics. These comparisons show that "hard to detect" must name a score family and representation. They do not establish RF permutation as a universally superior detector, because the method comparison lacks a frozen simultaneous paired interval and the best-per-mechanism choice uses oracle labels retrospectively.

At the eleven-mechanism level, D and X have a positive descriptive rank association, but category membership explains much of the apparent relation and leave-one-mechanism-out checks do not support a portable prediction rule. This is why Fig. 2 is a profile map rather than a fitted calibration curve. The controlled instantiation can identify mechanisms for which a diagnostic and a learner co-vary; it cannot infer exploitability for an unseen mechanism from D alone.

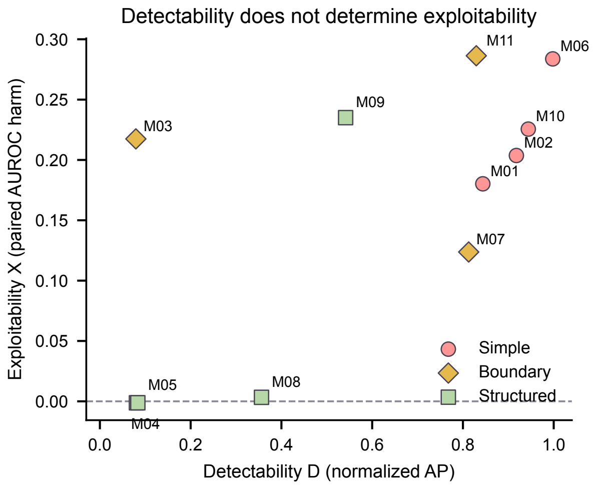

Figure 2: CDX measurement profiles. Each point is one mechanism mean over the frozen registry. D is normalized average precision under the primary training-side mutual-information diagnostic; X is paired AUROC harm. Shape and color encode construction family. The figure is descriptive and does not fit a deployment rule.

4.3 Natural Boundaries Broaden Context, Not the Sampling Frame

Five selected real-data case studies provide boundary-specific consistency checks (Table 3): UCI Bank Marketing [20], a local Lending Club mirror, U.S. flight delays [21], Chicago food inspections [2], and NYC 311 [3]. All have positive mean paired harm, but their size, feature representation, and contamination labels differ. The mean over these five selected tasks is 0.273. We do not use these cases for population inference.

The five case means span a wide range rather than forming a homogeneous replication set. Flight and inspection boundaries show large observed harm, whereas the NYC 311 boundary is close to the strict reference. This spread is substantively useful: it shows that declaring a field unavailable does not determine the magnitude of model exploitation. However, the tasks differ in sample construction, table width, encoded representation, unavailable-field count, and application semantics. Averaging them provides a compact audit summary, not a transport estimate. The fact that all five observed means are positive is reported as a fixed-case description and is not promoted to a population claim.

The matched-cost governance sensitivity is also heterogeneous. The descriptive mean P3-minus-mean-P2 SDR across the five cases is +0.187 (case-bootstrap interval [+0.025,+0.323]), but the five cases do not form a population sample. Bank Marketing (+0.102), Lending Club (+0.266), BTS Flights (+0.348), and Chicago Food (+0.326) favor P3, whereas NYC311 favors P2 (-0.108). The exact two-sided sign-flip value over the five fixed case effects is 0.1875. We therefore classify natural governance as mixed descriptive evidence rather than external confirmation.

NYC311 explains why the positive four-of-five count is not a transport claim. Its signed harm is +0.013, but the symmetric repair opportunity is only 0.019. At $k = 8$ of 40 fields, hash-verified

Table 1: CDX measurement profiles for the frozen mechanism registry. X is permissive-minus-strict test AUROC; D is normalized average precision for the primary training-side MI ranker. Intervals are 95% task-hierarchical intervals. Mechanism rows are descriptive; the prespecified simple-minus-structured contrast appears in the text.

ID	Mechanism	Family	X	95% interval	D
M01	Direct Target Copy	simple	+0.180	[+0.164, +0.196]	0.844
M02	Noisy Target Proxy	simple	+0.204	[+0.186, +0.222]	0.918
M03	Nonlinear Target Transform	boundary	+0.217	[+0.201, +0.234]	0.079
M04	Post-Outcome Aggregation	structured	-0.001	[-0.004, +0.001]	0.081
M05	Temporal Look-Ahead	structured	-0.001	[-0.003, +0.001]	0.084
M06	Redundant Leakage Cluster	simple	+0.284	[+0.261, +0.306]	0.998
M07	Sparse Subgroup Leakage	boundary	+0.124	[+0.114, +0.133]	0.813
M08	Entity Leakage	structured	+0.003	[-0.002, +0.009]	0.356
M09	Source Leakage	structured	+0.235	[+0.216, +0.254]	0.541
M10	Mixed Leakage	simple	+0.226	[+0.206, +0.245]	0.944
M11	Graph-Mediated Leakage	boundary	+0.286	[+0.263, +0.309]	0.829

P3 selection removes `resolution_description` but misses `status`, giving 50.0% recall; it simultaneously removes 7 contract-valid context fields and retains 81.6% of legitimate fields. The resulting -0.108 advantage is therefore a low-opportunity failure in which deletion cost exceeds the available distortion. Here "contract-valid" means available under the declared complaint-creation boundary, not necessarily useful or stable for deployment.

5 R: Matched-Cost Repair Results

5.1 Blind MI Adds Repair Value Beyond Deletion Count

At a 20% encoded-column budget, P3 exceeded the within-key mean of 20 P2 realizations by +0.043 SDR for the primary LR panel (95% paired task-cluster interval [+0.004,+0.078]). The corroborative B2 extension gives +0.055 for RF ([+0.025, +0.082]) and +0.056 for LightGBM ([+0.028, +0.082]). Direct paired contrasts among these named implementations all cross zero, so no reliable difference is detected; this is not equivalence, learner invariance, or baseline continuity with the measurement matrix. The encoded-column budget curve additionally shows the recall–retention trade-off across 1%, 5%, 10%, and 20% removal.

Random removal is a meaningful negative control in this setting because feature deletion can improve SDR without locating contamination. A permissive model may overfit weak legitimate fields, and deleting any such field can move its AUROC toward the strict score. P3 must therefore exceed the within-key mean P2 result, not merely obtain positive SDR. The observed advantage at all four nominal budgets indicates that MI ranking contributes information beyond deletion count in this LR panel.

The budget curve should not be interpreted as four fully independent policy experiments. On narrow tasks, adjacent fractions can map to the same rounded integer $k_q(p)$ and therefore reuse one fitted result under multiple nominal labels. The 5,500 task keys are complete at each nominal fraction, but the number of distinct interventions varies by task width. Figure 3 accordingly summarizes operational budget labels; it does not fit a smooth dose-response model.

5.2 Repair Opportunity Resolves the Structured Average

At a 20% budget, Δ_R was +0.094 for simple mechanisms (CI [+0.050, +0.133]) and +0.039 for boundary mechanisms (CI [-0.000, +0.073]). For structured mechanisms it was -0.004 (CI [-0.039, +0.026]). The structured interval crosses zero, so this experiment finds no reliable P3 advantage for that family. It does not establish equivalence or a general failure of MI on structured leakage. M04, M05, and M08 have little initial strict–full gap, whereas M09 combines an LR opportunity of 0.225 with +0.149 repair advantage (CI [+0.126,+0.171]) and remains positive for RF and LightGBM.

The opportunity strata sharpen this interpretation. The lowest LR gap quartile has $\Delta_R = -0.056$ (CI [-0.096,-0.021]), whereas the highest quartile has +0.203 ([+0.179,+0.223]). The sparse archetype is a separate negative regime (-0.118, [-0.160, -0.093]); the other four archetypes are positive, and every leave-one-archetype-out point estimate remains positive. These are sensitivity patterns, not evidence that opportunity or archetype causally determines repair.

We diagnosed the sparse failure without re-fitting a downstream model. A post-hoc reconstruction matched all 1,100 P3 selection hashes to the frozen B1 ledger. The mean is negative in 4/4 tasks and 9/11 mechanisms; P3 recall is 57.0% and legitimate retention is 84.8%. The clean sparse generator concentrates its legitimate signal in $1.1x_0 - 0.8x_3 + 0.5\mathbf{1}[x_5 > 0]$. At 20%, P3 removes on average 1.92 of these three fields; x_0 and x_3 are removed in 89.3% and 84.9% of keys. Thus the failure is consistent with a marginal ranker targeting concentrated legitimate signal, not merely with collapsed contamination recall. Because signal concentration was not separately randomized, this is a construction-grounded diagnosis rather than a causal decomposition.

The family decomposition is a direct effect comparison, not an inference from one interval being above zero and another crossing zero. Each family estimate recomputes the paired P3–P2 difference within its declared mechanism set and then clusters over tasks. The positive simple and boundary estimates therefore identify where the aggregate gain is concentrated. The structured estimate instead averages mechanisms with different repair opportunity and should not be treated as a category-level capability boundary.

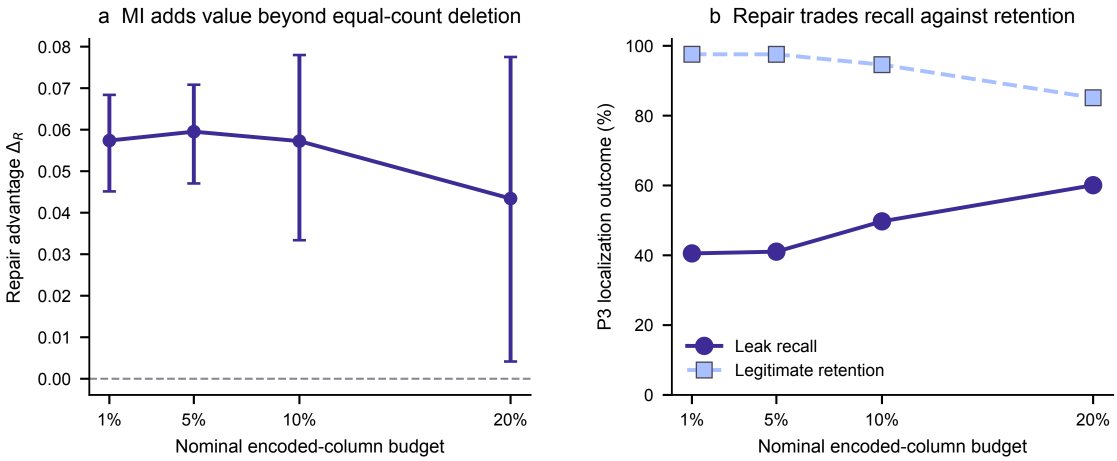

Figure 3: Budget sensitivity of the R layer. a, observed repair advantage Δ_R with paired task-cluster 95% intervals over 20 controlled tasks. Positive values favor P3 over mean P2. b, contaminated-field recall and legitimate-field retention for P3. P2 and P3 remove the same encoded cost at every task and nominal budget.

Table 2: CDXR repair stress tests at the 20% encoded-column budget. Δ_R is P3 minus the within-key mean of 20 matched P2 realizations; positive values favor blind MI. Intervals resample the 20 controlled tasks, except archetype rows, which contain four tasks. Opportunity strata are sensitivity analyses, not causal effects.

Stress test	Scope	Learner	Opportunity	Δ_R	95% interval	Unit
Across learners	All mechanisms	LR	–	+0.043	[+0.004, +0.078]	5500 keys
Across learners	All mechanisms	RF	–	+0.055	[+0.025, +0.082]	5500 keys
Across learners	All mechanisms	LightGBM	–	+0.056	[+0.028, +0.082]	5500 keys
Mechanism family	simple	LR	–	+0.094	[+0.050, +0.133]	20 tasks
Mechanism family	boundary	LR	–	+0.039	[-0.000, +0.073]	20 tasks
Mechanism family	structured	LR	–	-0.004	[-0.039, +0.026]	20 tasks
Counterexample	M09	LR	0.225	+0.149	[+0.126, +0.171]	500 keys
Opportunity	Q1_low	LR	[0.0000,0.0028]	-0.056	[-0.096, -0.021]	1375 keys
Opportunity	Q4_high	LR	[0.2478,0.4296]	+0.203	[+0.179, +0.223]	1375 keys
Archetype	sparse	LR	–	-0.118	[-0.160, -0.093]	4 tasks

5.3 The Cost Contract Changes Aggregate Support

The primary protocol charges encoded columns, so removing the complete M09 source field costs eight units. A frozen sensitivity instead treats its eight one-hot indicators as one semantic group, scores each group by the maximum training-side MI of its members, and samples P2 uniformly over groups. M09 is the only mechanism whose frozen representation expands one semantic source into several encoded columns; the remaining 5,000 keys therefore retain their encoded-cost results in the 5,500-key recomposition.

The two cost contracts give

$$\widehat{\Delta}_{R,\mathrm{enc}}^{\mathrm{M09}} = +0.149, \qquad \mathrm{CI}_{.95} = [+0.126, +0.171], \tag{28}$$
$$\widehat{\Delta}_{R,\mathrm{grp}}^{\mathrm{M09}} = +0.109, \qquad \mathrm{CI}_{.95} = [+0.054, +0.158]. \tag{29}$$

The paired group-minus-encoded change is -0.040; its 95% CI is [-0.076,-0.008]. After recomposition, the semantic-group overall estimate is +0.040. Its interval crosses zero ([-0.003,+0.077]), whereas the encoded-cost estimate is +0.043 with a positive interval. M09 is not an artifact of charging eight units, but the aggregate repair claim is sensitive to the representation-cost definition.

Table 3 places the cost contract beside the five natural boundaries. Their mean repair response is +0.187, but only four cases favor P3; NYC311 is negative. The five-case exact sign-flip value is 0.1875. Natural governance is therefore mixed despite its positive descriptive mean.

5.4 CDXR Connects Measurement to Action

The measurement and governance results align at the family level without establishing a universal transfer rule. Simple contamination is usually highly detectable, strongly exploited, and amenable to top-ranked field removal. Boundary mechanisms are heterogeneous but retain a positive aggregate P3 advantage. Structured mechanisms encode invalidity through time, entity, and source relationships that a univariate training-side score does not represent uniformly. This interpretation is consistent with the observed profiles and policy outcomes, but it is not a causal decomposition of policy performance.

The crossed profiles also warn against treating high D as sufficient for governance. M09 is detectable and harmful under its fixed representation, yet structured governance as a family does

Table 3: Context and cost-contract sensitivities. Natural rows report fixed-case harm (X) and LR repair response at 20%; no case-level inferential interval is claimed. Cost rows report controlled-registry Δ_R with 95% task-cluster intervals. Natural governance is mixed because NYC311 is negative.

Setting	Case / scope	Boundary or cost	X / opportunity	Δ_R	95% interval	Evidence
Natural	BankMarketing	Before call completion	+0.131	+0.102	–	fixed case
Natural	LendingClub	At loan origination	+0.319	+0.266	–	fixed case
Natural	BTSFlights	Before scheduled departure	+0.470	+0.348	–	fixed case
Natural	ChicagoFood	Before inspection outcome	+0.430	+0.326	–	fixed case
Natural	NYC311	At complaint creation	+0.013	-0.108	–	fixed case
Cost	All mechanisms	encoded column	–	+0.043	[+0.004, +0.078]	positive interval
Cost	All mechanisms	semantic group	–	+0.040	[-0.003, +0.077]	interval crosses 0
Cost	M09	semantic group	–	+0.109	[+0.054, +0.158]	positive interval

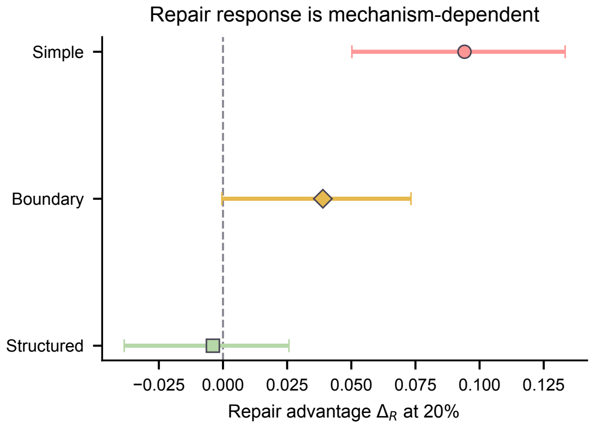

Figure 4: Mechanism-family heterogeneity at a 20% encoded cost. Points show Δ_R and bars show paired task-cluster 95% intervals. Marker shape and color redundantly encode family. The structured interval crosses zero and is not an equivalence result.

not show a reliable advantage. A field ranking can identify association while still removing legitimate correlates or failing to represent the semantic group that should be governed. CDXR therefore evaluates localization, downstream distance, recall, and retention as separate outcomes.

The policy results also separate identification from repair. A contaminated column counted in recall may be statistically redundant with another retained column, so removing it need not change AUROC. Conversely, removing a legitimate correlate can accidentally offset permissive inflation and improve SDR while reducing retention. The repair vector $R = (\mathrm{SDR}, \mathrm{Rec}, \mathrm{Ret})$ prevents either case from being reported as unqualified success. In the present results, low-budget P3 combines a positive paired advantage with high legitimate retention, whereas larger budgets buy additional recall at a visible retention cost.

6 Related Work

Data contracts and validation. Production ML creates data-management problems around understanding, validating, cleaning, and enriching training data [13]. Declarative quality systems scale constraints and anomaly checks over large datasets [18], while ML data validation additionally targets schemas, unexpected patterns, and training-serving skew [14]. CDXR is complementary: observed-data constraints cannot by themselves decide whether a field is semantically available at a prediction event, and anomaly detection does not measure whether a removal policy repairs a boundary-valid reference. CDXR adds that semantic and interventional contract.

Leakage semantics. Classical leakage formulations distinguish information that is available when a model is developed from information that will be available when its prediction is consumed [9]. This semantic perspective is stronger than a correlation rule: an aggressively predictive field may be legitimate, while a weak future-derived field remains invalid. Recent work links leakage to reproducibility in ML-based science [8] and formalizes domain-specific label leakage in health care [4]. Empirical studies in connectome prediction further show that leakage can inflate reported performance under otherwise familiar modeling pipelines [16]. CDXR adopts the same boundary-first principle, but extends it to a repair contract in which validity, localization, exploitation, opportunity, and intervention response remain separately auditable.

Detection at code, data, and representation levels. Static program analysis targets leakage pathways visible in data-science code, such as fitting preprocessing outside the training split [19]. Contamination audits in language benchmarks instead often ask whether evaluation content was memorized and subsequently exploited [11]. Feature-level screening occupies another level: it ranks columns from observed training data but typically cannot determine whether their information is temporally or operationally available. These approaches are complementary because they observe different evidence. Code analysis can reveal an invalid fit scope without ranking columns; a statistical ranker can reveal a target proxy without knowing its lineage; and a semantic contract can identify a future field even when its sample association is weak. The C/D/X interfaces of CDXR make these non-equivalences explicit, while the R interface tests whether a detector ranking remains useful after an equal-cost negative control.

Tabular evaluation under controlled shifts. TableShift studies tabular prediction under distribution shifts, while broad comparative benchmarks examine when boosted trees, bagged trees, linear models, and neural architectures succeed [5, 7, 12]. Those resources vary datasets, domains, or learner families to characterize predictive performance. LeakBench-Tab holds a task instance and split fixed while perturbing the information boundary through auditable mechanisms. Learner diversity is used to measure exploitability conditional on representation, not to declare a new winner among logistic regression, random forests, boosting, and TabM [1, 6, 10, 15]. This focus also explains why the natural cases are treated as boundary audits rather than merged with the controlled panel.

Feature selection versus leakage governance. Mutual information, linear coefficients, correlation, and permutation importance measure predictive relevance under different function classes. None is intrinsically a semantic validity test. Conventional feature selection usually optimizes predictive performance or sparsity; leakage governance asks whether a constrained intervention moves an evaluation toward a boundary-valid reference while preserving legitimate information. Our P2/P3 comparison holds removal cost fixed. It reports leak recall, legitimate retention, and downstream strict distance. The method contribution is the contract-grounded repair architecture and its controls; MI supplies a deliberately conventional test policy.

7 Discussion

7.1 What CDXR Changes

Leakage audits often end after declaring suspicious fields or measuring an inflated score. Our results show that an actionable evaluation needs at least four linked but distinct objects: the semantic contamination label, a blind localization score, downstream exploitability, and the effect of a removal policy under explicit cost. Reporting only one object hides important cases. Low D does not validate a field, high X does not reveal which field caused the inflation, and high D does not guarantee that removal will preserve legitimate utility.

Matched-cost evaluation is central to this conclusion. P3 is not compared with a baseline that removes fewer fields, and the evaluation records retention as well as recall. The 1% result is practically informative because it combines a positive distance reduction with high legitimate retention. The 20% result shows the opposite side of the trade-off: recall rises, but retention falls.

7.2 Implications for Leakage Audits

The findings suggest an audit sequence rather than an automatic deletion rule. An analyst should first declare the prediction event and label each field by availability or lineage. Blind scores can then prioritize manual review, but their rankings should be evaluated against a held-back semantic mask when a benchmark is available. Finally, any removal policy should be compared with an equal-cost negative control and assessed on downstream behavior. This sequence prevents a high association score from silently becoming a validity judgment.

Budget choice is part of the governance decision. At small budgets, MI removal preserves most legitimate fields and can cheaply eliminate direct proxies. At larger budgets, the marginal recall gain is purchased with additional legitimate-field deletion. A deployment process would therefore need a domain-specific cost function over missed contamination, lost legitimate signal, calibration, and decision utility. The benchmark exposes ingredients for that choice but deliberately does not collapse them into a universal scalar utility.

The semantic-group sensitivity shows why cost definition must be part of the audit contract. Grouping M09's indicators preserves its positive effect but widens the full-panel uncertainty enough to cross zero. Richer policies could add feature-lineage constraints or treat temporal availability and entity aggregation as first-class metadata. Their evaluation should preserve the same oracle-isolation rule: metadata available to the policy may be used, but the hidden contamination label should remain an evaluation object.

7.3 Alternative Explanations and Required Follow-Ups

The observed family pattern can have several compatible explanations. Simple mechanisms may be easier for MI to rank because their invalid signal is carried by individual encoded columns. Structured mechanisms may require group or conditional statistics, but they may also exhibit weaker exploitability in the particular generators, leaving less inflation for any policy to repair. Learner representation adds a third possibility, although the similar LR, RF, and LightGBM contrasts do not reveal a detectable learner interaction in the present panel. The current factorial design describes the joint outcome of these factors; it does not identify their separate causal contributions.

The largest remaining ambiguity concerns policies that use lineage, temporal, or entity metadata rather than blind univariate association. These policies would test whether low-gap structured mechanisms become governable when the intervention represents their construction semantics. Natural evaluation also remains narrow: five fixed cases cannot establish a population-level transport claim. The NYC311 anatomy explains the observed deletion path under its declared contract, but application-specific review must still decide whether fields such as identifiers and service descriptors belong in an operational policy.

7.4 Limitations

The controlled tasks are synthetic panel generators, not twenty independent real-world application environments. Their bootstrap intervals describe registry stability, not a task population. The governance experiment covers LR, RF, and LightGBM, but these named implementations do not establish invariance across model families or tuning choices. The direct learner-interaction intervals cross zero; they are tests of detected difference, not equivalence tests. The RF and LightGBM extension uses separately frozen 100-estimator repair factories and refitted strict/full baselines under a disclosed protocol deviation. Those results retain within-run paired validity but not model-factory or bitwise baseline continuity with the earlier measurement run.

P2 uses 20 frozen random field selections for each task key and budget, and the primary estimand averages P2 within key before task-cluster resampling. This integrates the declared random-policy variance but does not cover every possible randomization scheme. Nominal budget labels can also map to the same integer removal count on tasks with few fields, so adjacent budget points are not fully distinct interventions for every task.

The primary blind policy is univariate mutual information. It does not encode lineage, temporal availability, entity grouping, or source semantics. The structured-family average should not be generalized to all metadata-aware or group-aware policies. Conversely, P3's aggregate advantage should not be read as evidence that association determines validity.

The five real-data audits are selected, boundary-specific case studies. One local Lending Club mirror cannot be redistributed until its source and license are verified. AUROC distance does not replace calibration, decision utility, or operational monitoring. Runtime and resource overhead are also outside the current frozen evidence and should be added before making systems-efficiency claims.

7.5 Threats to Validity

Construct validity. SDR measures movement toward a frozen strict reference, not the business value of a deployed decision. Its interpretation depends on the correctness of each declared prediction boundary and mechanism contract. Recall and retention use the contamination mask only after policy selection and are therefore evaluation measures rather than deployable observables. Encoded-column recall also differs from semantic-field recall when categorical fields expand into multiple indicators. CDXR states the encoded unit explicitly, but a production policy may require group-level metrics.

Internal validity. The paired protocol holds splits, task content, model seeds, and removal cost fixed within each comparison. Bundle, split, and runner hashes reduce implementation drift, but they cannot rule out all library-specific behavior. Hyperparameters are fixed named implementations rather than optimized independently for every mechanism; this improves comparability but may understate the response of a specially tuned learner. Protocol amendments are provenance-bound and replace affected cells, yet their necessity also illustrates how semantic bugs can survive ordinary completion checks.

External and statistical validity. The designed registry deliberately varies mechanism structure, but it does not define a sampling frame of tabular applications. Mechanism-family intervals and bootstrap probabilities should be read as stability summaries for this registry. The natural cases broaden the contexts inspected but remain selected case studies; their governance result is mixed and descriptive. Multiplicity correction protects the declared contrast families, but descriptive mechanism profiles and diagnostic sensitivity rows should not be mined as independent confirmatory discoveries.

8 Reproducibility and Artifact Boundary

The release binds the canonical 27,500-cell measurement table, separate measurement and governance claim states, repair runners, the 709,500-row revision ledger, bootstrap analyses, and paper-facing CSV tables by SHA-256. The release validator deep-runs statistical tables and reports 233 passing tests with no failures. Superseded runners and intermediate cluster summaries are retained for provenance but excluded from claim generation. Paper figures and tables are regenerated from three compact CSV assets: the mechanism results, governance results, and natural cases.

Reproduction is organized in three layers. The *paper layer* regenerates all numeric macros, tables, and figures from the three compact CSV assets and records their hashes in a paper-artifact manifest. The *analysis layer* rebuilds measurement and governance summaries from canonical cell ledgers, including the task-level bootstrap objects. The *execution layer* reruns frozen task bundles through named model and policy runners after verifying bundle, split, configuration, and code identities. A reader can therefore inspect the paper without rerunning models, recompute claims without rebuilding tasks, or execute the complete experiment when the required packages and hardware are available.

Claim state is machine-readable and distinct from prose. Each central claim records an estimand, evidence tier, allowed wording, prohibited wording, and scope limitation. This is the artifact realization of certificate Γ: invalid access, cost, pairing, or provenance blocks generation before statistical support is considered. The post-hoc failure anatomy has a separate manifest and cannot promote the confirmatory claim state. The paper generator does not infer support from arbitrary tables; it consumes the frozen paper-facing assets. This separation prevents a later figure edit, exploratory table, or superseded run from silently changing a conclusion.

The public artifact and reproduction instructions are maintained at https://github.com/lzzzhh/LeakBench-Tab. The paper-facing artifact does not redistribute the local-only Lending Club mirror. All non-oracle policy selectors can be audited without access to the contamination mask; the mask is used only for evaluation after selection.

9 Conclusion

CDXR reframes tabular leakage repair as a contract-grounded evaluation architecture rather than a detector leaderboard. C freezes semantic invalidity, D measures blind localization, X measures learner exploitation, and R evaluates the intervention against equal-cost deletion. Semantic precedence, oracle isolation, matched cost, paired reference evaluation, and task-level uncertainty make the resulting claim falsifiable.

The LeakBench-Tab instantiation supports one bounded repair result: under the primary 20% encoded-column contract, training-side MI has a positive advantage for LR. Separately frozen, protocol-deviating RF and LightGBM extensions corroborate its direction without establishing implementation continuity or learner invariance. CDXR also makes the boundary visible. Low-opportunity strata and the sparse archetype are negative; M09 refutes a blanket structured-failure narrative; semantic grouping moves the aggregate interval across zero; and natural governance is mixed. The contribution is therefore not that MI solves leakage, but that leakage repair can be measured without conflating validity, detectability, exploitation, and deletion cost. Lineage-, time-, and entity-aware policies can now be tested through the same interfaces.

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